Double pendulum and chaotic motion

NAVY 2019

What is double pendulum?

- Simple physical system, where we can observe the chaotic behavior (chaotic motion)
- Components:
  - 2 pendulums with weights m_1 and m_2
  - 2 massless wires with lengths l_1 and l_2
  - Both pendulums are deviated – angles: θ_1 and θ_2

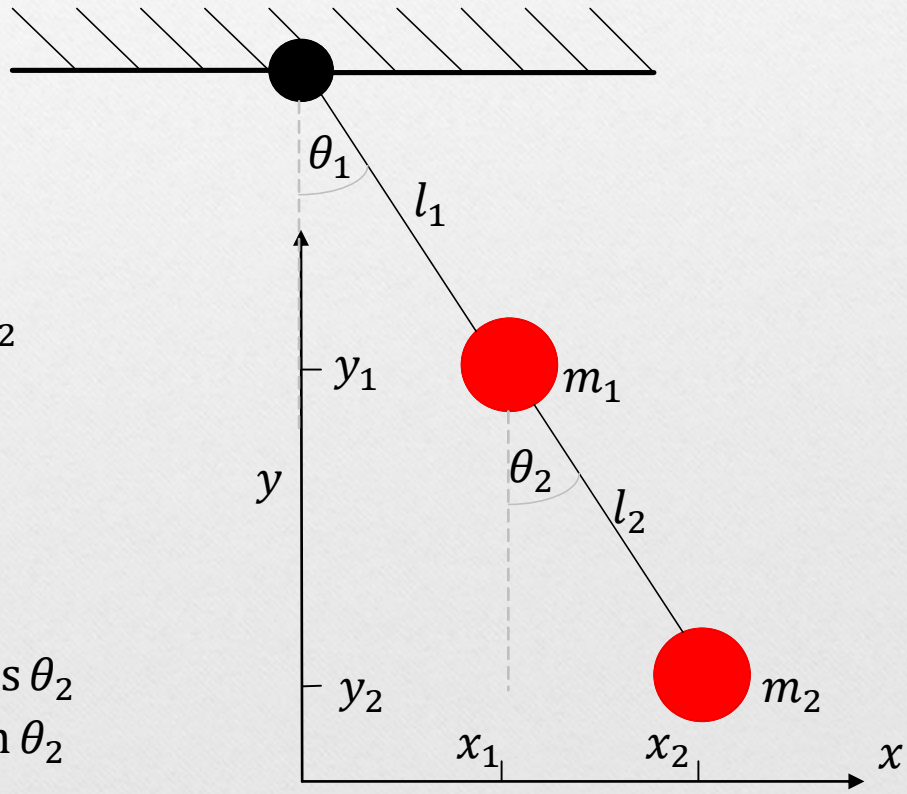
Positions of pendula

Positions:

- $x_1 = l_1 sin\ \theta_1$
- $y_1 = -l_1 \cos\theta_1$
- $x_2 = l_1 sin\theta_1 + l_2 \sin\theta_2$
- $y_2 = -l_1 \cos\theta_1 - l_2 \cos\theta_2$

Velocities:

- $\dot{x}_1 = l_1\dot{\theta}_1 \cos\theta_1$
- $\dot{y}_1 = l_1\dot{\theta}_1 \sin\theta_1$
- $\dot{x}_2 = l_1\dot{\theta}_1 \cos\theta_1 + l_2\dot{\theta}_2 \cos\theta_2$
- $\dot{x}_y = l_1\dot{\theta}_1 \sin\theta_1 + l_2\dot{\theta}_2 \sin\theta_2$



Potential and kinetic energy, Lagrangian

Potential energy: $\boldsymbol{V = mhg}$

$$V = m_1 g y_1 + m_2 g y_2 = -(m_1 + m_2) l_1 g cos\ \theta_1 - m_2 l_2 g cos\ \theta_2$$

Kinetic energy: $\boldsymbol{T = \frac{1}{2}\mathrm{mv}^2}$

$$T = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 = \frac{1}{2} m_1 (\dot{x}_1^2 + \dot{y}_1^2) + \frac{1}{2} m_2 (\dot{x}_2^2 + \dot{y}_2^2)$$

$$= \frac{1}{2} m_1 l_1^2 \dot{\theta}_1^2 + \frac{1}{2} m_2 [l_1^2 \dot{\theta}_1^2 + l_2^2 \dot{\theta}_2^2 + 2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2)]$$

Lagrangian: $\boldsymbol{\mathcal{L} = T - V}$

$$\mathcal{L} = \frac{1}{2}(m_1 + m_2) l_1^2 \dot{\theta}_1^2 + \frac{1}{2} m_2 l_2^2 \dot{\theta}_2^2 + m_1 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2) +$$

$$(m_1 + m_2) l_1 g\ cos\theta_1 + m_2 g l_2 cos\theta_2$$

Euler-Lagrange equations

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i}\right) - \frac{\partial \mathcal{L}}{\partial q_i} = 0 \qquad \text{for } q_i = \theta_1, \theta_2$$

We will get:

$$m_2 l_2 \ddot{\theta}_2 + m_2 l_1 \ddot{\theta}_1 \cos(\theta_1 - \theta_2) - m_2 l_1 \dot{\theta}_1^2 \sin(\theta_1 - \theta_2) + m_2 g sin\, \theta_2 = 0$$

Expressions for acceleration

Acceleration of the 1st and 2nd pendulum:

$$\ddot{\theta}_1 = \frac{m_2 g \sin\theta_2 \cos(\theta_1 - \theta_2) - m_2 \sin(\theta_1 - \theta_2)\left[l_1 \dot{\theta}_1^2 \cos(\theta_1 - \theta_2) + l_2 \dot{\theta}_2^2\right] - (m_1 + m_2) g sin\, \theta_1}{l_1[m_1 + m_2 \sin^2(\theta_1 - \theta_2)]}$$

$$\ddot{\theta}_2 = \frac{(m_1 + m_2)\left[l_1 \dot{\theta}_1^2 \sin(\theta_1 - \theta_2) - g sin\, \theta_2 + g sin \theta_1 \cos(\theta_1 - \theta_2)\right] + m_2 l_2 \dot{\theta}_2^2 \sin(\theta_1 - \theta_2)\cos(\theta_1 - \theta_2)}{l_2[m_1 + m_2 \sin^2(\theta_1 - \theta_2)]}$$

Application

- Set angles θ_1 and θ_2, for example $\theta_1 = \frac{2\pi}{6}$ and $\theta_2 = \frac{5\pi}{8}$
- At the beginning, the angle velocity is 0 ⇒ $\dot{\theta}_1 = \dot{\theta}_2 = 0$
- Use `odeint` function solving the differential equations

- Python: `od=odeint(get_derivative, state_0, t, args=(`l_1, l_2, m_1, m_2`))`
  where `get_derivative` calculates the accelerations $\ddot{\theta}_1$ and $\ddot{\theta}_2$

  - The function `get_derivative` will return tuple:
    - $\dot{\theta}_1, \ddot{\theta}_1, \dot{\theta}_2, \ddot{\theta}_2$
  - t is the array of time steps ⇒ function `odeint` will return result for each time step ⇒ array

  - We can get the new angles $\theta_1 =$ `od[:,0]`, $\theta_2 =$ `od[:,2]`

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  - We can get the new angles $\theta_1 =$ `od[:,0]`, $\theta_2 =$ `od[:,2]`
  - From the new angles, we can calculate new positions of the pendula: x_1, y_1, x_2, y_2

Literature

- https://scipython.com/blog/the-double-pendulum/